Final Examination Practice Sheet

Course Code: INT396
Course Title: UNSUPERVISED LEARNING
Section(s): K3P24MK, K3P24WV
Syllabus Covered: Unit 1 to Unit 6
Pattern: per unit, 8 MCQs (1 mark) and 2 long questions (10 marks)
Purpose: self-practice for the end-term examination

Instructions: attempt every question without notes, then check your answers with your instructor. For the long questions, write structured answers in the form you would use in the examination.

Unit 1: Foundations of Unsupervised Learning

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. Two delivery points are compared by adding the absolute differences of their east-west and north-south positions, without squaring them. Which measure is being used?

(A) Euclidean distance
(B) Manhattan distance
(C) Cosine similarity
(D) Jaccard similarity

2. A hospital dataset of patient records has no column stating which disease each patient has. Which element that supervised learning requires is missing?

(A) The feature columns
(B) The number of records
(C) The unit of each measurement
(D) A label column giving the correct answer for each record

3. Two customers are described by purchase counts, all of which are zero or more. In which range must the cosine similarity between them lie?

(A) 0 to 1
(B) -1 to 0
(C) 0 to 100
(D) minus infinity to plus infinity

4. The cosine similarity between the spending vectors of two shoppers is exactly 1.0. What does this indicate?

(A) They spend the same total amount
(B) They buy no item in common
(C) They spend in the same proportions across categories, even if their totals differ
(D) Their Euclidean distance is zero

5. An online wallet flags the few logins whose time, device and location are unlike anything the account has produced before. Which unsupervised task is this?

(A) Market segmentation
(B) Association rule mining
(C) Anomaly detection
(D) Dimensionality reduction

6. A customer table holds annual income in rupees and age in years. If Euclidean distance is computed without scaling, what happens?

(A) Income dominates the distance
(B) Age dominates the distance
(C) Both features contribute equally
(D) The distance becomes negative

7. A school clusters its students by study habits, and no grades or categories were ever recorded. A teacher asks for the "accuracy" of the clustering. Why can it not be given?

(A) Clustering is always 100 per cent accurate
(B) There are too many students
(C) Accuracy applies only to regression
(D) There are no true labels to compare the clusters against

8. A laboratory has confirmed diagnoses for 200 of its 20,000 scans and trains one model on all 20,000 scans. Which learning paradigm is this?

(A) Supervised learning
(B) Semi-supervised learning
(C) Unsupervised learning
(D) Reinforcement learning

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. A video-streaming service represents each user as a vector of minutes watched in each genre. (10 Marks)

(a) Explain what Euclidean distance, Manhattan distance and cosine similarity each measure about a pair of users. *[6]*

(b) Two users watch the same mix of genres, but one watches three times as many minutes in total. State which of the three measures treats them as most alike, and give the reason. *[4]*

10. Distinguish supervised, unsupervised and semi-supervised learning, using an e-commerce company as the setting. (10 Marks)

(a) For each of the three, describe the data it requires and the question it answers. *[6]*

(b) Give one e-commerce application of each. *[4]*

Unit 2: Partition-Based Clustering

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A k-Means cluster holds the points (1, 5), (3, 9) and (8, 7). What is the new centroid after the update step?

(A) (3, 7) (B) (4, 6)

(C) (4, 7) (D) (6, 7)

2. A one-dimensional cluster holds the values 4, 10 and 16. What are its centroid and its contribution to the within-cluster sum of squares?

(A) Centroid 10, contribution 72 (B) Centroid 10, contribution 12

(C) Centroid 10, contribution 36 (D) Centroid 12, contribution 72

3. Using the k-Medoids cost (sum of absolute distances), which record is the medoid of the cluster 3, 8 and 20?

(A) 3, with cost 22 (B) 8, with cost 17

(C) 10.33, with cost 17 (D) 20, with cost 29

4. In k-Means++, the first centre has been chosen and two candidate points lie at distances 3 and 4 from it. What is the probability that the point at distance 4 is chosen next?

(A) 0.57 (B) 0.43

(C) 0.36 (D) 0.64

5. Exam scores have mean 50 and standard deviation 4. Ravi scored 44. What is his score after standardisation (z-score)?

(A) -1.5 (B) 1.5

(C) -6 (D) -0.67

6. Inertia readings are: k = 1 gives 1,200, k = 2 gives 600, k = 3 gives 250, k = 4 gives 230 and k = 5 gives 220. Where is the elbow?

(A) k = 1 (B) k = 2

(C) k = 3 (D) k = 5

7. A cluster of house prices around 50 lakh gains one mansion priced at 20 crore. What happens to the k-Means centroid of that cluster?

(A) It does not change (B) It moves to the median of the cluster

(C) It becomes the mansion itself (D) It is pulled towards the mansion, because the centroid is a mean

8. An analyst notices that randomly chosen starting centres often land close together and give poor clusters. Which k-Means variant was designed to fix exactly this?

(A) MiniBatch k-Means (B) k-Means++

(C) k-Medoids (D) Fuzzy c-means

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. A gym runs k-Means on its members using the number of visits per month (0 to 30) and the annual membership fee in rupees (5,000 to 60,000), without scaling either column. (10 Marks)

(a) Analyze which of the two features decides the clusters, and give the reason. *[5]*

(b) Infer how the clusters would change after both columns are standardised. *[5]*

10. A retailer runs k-Means for k = 2 to 7. The inertia falls steeply up to k = 4 and then flattens, and the silhouette score is highest at k = 4. (10 Marks)

(a) Analyze what inertia and the silhouette score each tell you about a clustering. *[5]*

(b) Infer the value of k you would choose, give the reason, and explain why inertia alone cannot be used to pick k. *[5]*

Unit 3: Hierarchical and Density-Based Clustering

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. You want DBSCAN to treat points up to 2 km apart as neighbours. Which parameter do you set to 2 km?

(A) MinPts (B) k

(C) The linkage criterion (D) eps

2. With MinPts set to 5, a point has exactly 3 other points within distance eps. What is its status?

(A) It is a core point (B) It is not a core point; it may still be a border point if it lies within eps of a core point

(C) It is always noise (D) It is a cluster centre

3. Agglomerative clustering starts with 20 separate records. How many merges are needed before one cluster remains?

(A) 19 (B) 20

(C) 10 (D) 21

4. The pairwise distances between the members of two clusters are 4, 11, 6 and 8. What is their complete-linkage distance?

(A) 4 (B) 7.25

(C) 11 (D) 6

5. Two clusters hold 2 and 4 records, and the 8 cross-cluster pairwise distances sum to 48. What is the average-linkage distance?

(A) 8 (B) 6

(C) 12 (D) 4.8

6. A DBSCAN run labels almost every record as noise. What is the most likely cause?

(A) eps is too large (B) Too many clusters were requested

(C) The data was standardised (D) eps is too small or MinPts is too large

7. Two groups of houses are merged as soon as any one house in the first group is close to any one house in the second. Which linkage rule behaves this way?

(A) Complete linkage (B) Average linkage

(C) Single linkage (D) Ward's method

8. Why does agglomerative clustering become impractical on very large datasets?

(A) It needs all pairwise distances, so cost grows with the square of the number of records (B) It cannot handle numeric data

(C) It requires k in advance (D) It only works in two dimensions

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. An analyst runs k-Means with k = 2 on GPS points that form two crescent-shaped (half-moon) groups, and each resulting cluster cuts across both crescents. (10 Marks)

(a) Explain why k-Means cannot find these two groups. *[5]*

(b) State which method you would use instead, and what that method needs to be set before it runs. *[5]*

10. Compare single linkage, complete linkage and average linkage for clustering shop locations along a highway. (10 Marks)

(a) State how each of the three measures the distance between two clusters. *[5]*

(b) State which one gives long chain-like clusters and which gives compact ones, with the reason, and which is most affected by outliers. *[5]*

Unit 4: Dimensionality Reduction and Representation Learning

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. Principal components explain 45, 25, 15, 10 and 5 per cent of the variance. What is the smallest number of components that retains at least 80 per cent?

(A) 3 (B) 2

(C) 4 (D) 5

2. A survey with 120 questions is summarised by 30 principal components. By what percentage has the number of variables been cut?

(A) 25 per cent (B) 30 per cent

(C) 75 per cent (D) 90 per cent

3. The first two components explain 40 and 25 per cent of the variance, and the first three together explain 80 per cent. How much does the third component explain?

(A) 25 per cent (B) 15 per cent

(C) 20 per cent (D) 80 per cent

4. If you could keep only one new axis to describe a cloud of data points while losing as little information as possible, which axis would PCA choose?

(A) The feature with the largest mean (B) A randomly chosen axis

(C) The direction of least variance (D) The direction of greatest variance

5. An analyst applies PCA to a table holding weight in grams and temperature in degrees Celsius without standardising. What goes wrong?

(A) Temperature dominates the components (B) Nothing changes

(C) Weight dominates the components because of its larger raw variance (D) PCA fails to run

6. Two clusters appear far apart on a t-SNE plot. What can safely be concluded about the distance between the two groups?

(A) Little: t-SNE preserves local neighbourhoods, not large distances (B) The groups are very different

(C) The data contains exactly two clusters (D) The plot distance is in the original units

7. Which quantity best measures the quality of an autoencoder's compression?

(A) The number of layers (B) The training time

(C) The size of the bottleneck (D) The reconstruction error

8. The reduced features must be explained to auditors who need to know what each new feature means. Which technique is most appropriate?

(A) t-SNE (B) PCA, since each component is an explicit weighted sum of the original features

(C) An autoencoder (D) UMAP

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Assess the need for dimensionality reduction on a bank dataset holding 10,000 customers described by 1,200 features. (10 Marks)

(a) Evaluate two benefits it could deliver and one risk it introduces. *[5]*

(b) Judge whether the features must be standardised before PCA is applied, explaining your reasoning. *[5]*

10. Critique the claim in a project report: 'Our autoencoder compresses 300 sensor features into 15, and since the reconstruction error is very low, the 15 features contain all the useful information.' (10 Marks)

(a) Assess what a low reconstruction error does and does not establish. *[5]*

(b) Recommend the evidence the report should have presented instead. *[5]*

Unit 5: Association Rule Mining and Anomaly Detection

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A stationery shop records 800 transactions, and the itemset {pen, notebook} appears in 120 of them. What is its support?

(A) 0.12 (B) 0.15

(C) 0.20 (D) 0.67

2. The rule {coffee} -> {sugar} has support 0.10. Coffee appears in 0.25 of transactions and sugar in 0.20. What is the lift of the rule?

(A) 0.4 (B) 0.5

(C) 1.0 (D) 2.0

3. A rule has confidence 0.90 and lift 1.01. How should it be read?

(A) The consequent is common anyway, so the rule (B) It is a strong and useful rule

reveals little

(C) It shows a negative association

(D) Its support must be high

4. Apriori never counts {milk, bread, jam}, because {milk, jam} has already been found to be rare. Which principle allows this shortcut?

(A) Every superset of a frequent itemset is frequent

(B) Every rule must have lift above 1

(C) If an itemset is infrequent, every larger itemset containing it is also infrequent

(D) Itemsets must contain exactly two items

5. A fraud tool grows many random trees and flags the transactions that end up alone after only a few random cuts. Which method is this?

(A) Local Outlier Factor

(B) k-Means

(C) DBSCAN

(D) Isolation Forest

6. A card transaction sits in a sparse area, while its nearest neighbours sit in dense areas, so the Local Outlier Factor scores it as an outlier. What is LOF comparing?

(A) Its distance to the global centroid

(B) Its local density with the local densities of its neighbours

(C) The number of random splits needed to isolate it

(D) The support of its items

7. A hospital system screens prescriptions for dangerous drug interactions. Which error is more costly?

(A) A missed interaction (false negative), because the patient is harmed

(B) A false alarm, because a pharmacist must check it

(C) Both cost the same

(D) Neither, since doctors check everything

8. Raising the minimum support threshold from 0.02 to 0.06 cuts the frequent itemsets from about 5,000 to 9. What is the correct conclusion?

(A) The algorithm has failed

(B) The data has no patterns

(C) Most itemsets are now excluded, and some genuine but rarer patterns may be lost

(D) The confidence of every rule has risen

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. A bookshop mined association rules from 2,000 bills. The rule {notebook} -> {pen} has support 0.18, confidence 0.90 and lift 1.05. (10 Marks)

(a) State what support, confidence and lift each measure about a rule. *[5]*

(b) Explain why this rule, despite its high confidence, may be of little use, and state what a lift close to one means. *[5]*

10. A government audit office must flag suspicious claims among 2 million health-insurance claims, and must explain every flag to its investigators. (10 Marks)

(a) Describe how Isolation Forest and Local Outlier Factor each decide that a claim is unusual. *[5]*

(b) Recommend one of the two on the basis of speed and explainability, with your reason. *[5]*

Unit 6: Evaluation and Applications of Unsupervised Learning

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A record has mean distance 3 to its own cluster and mean distance 9 to the nearest other cluster. What is its silhouette value?

(A) 0.33

(B) 0.50

(C) 0.67

(D) 3.0

2. A record has mean distance 8 to its own cluster and 6 to the nearest other cluster. What is its silhouette value, and what does it indicate?

(A) -0.25; it is probably in the wrong cluster

(B) 0.25; it is well clustered

(C) 0.75; it is well clustered

(D) -2; the value is impossible

3. A student reports a silhouette coefficient of 1.4 for one record. Why must this be a mistake?

(A) Because the silhouette lies between 0 and 1

(B) Because it can never exceed 0.5

(C) Because negative values are impossible

(D) Because the silhouette always lies between -1 and +1

4. A record has a silhouette coefficient close to zero. What does this indicate?

(A) It lies deep inside its cluster

(B) It lies on the boundary between two clusters

(C) It is noise

(D) It is certainly misassigned

5. The Adjusted Rand Index needs known labels, while the silhouette needs only the data and the clusters. Which distinction does this illustrate?

(A) Supervised versus unsupervised learning

(B) Internal versus external validation measures

(C) Hard versus soft clustering

(D) Training versus testing

6. A customer segmentation is rebuilt on five random samples of the same customer base and gives noticeably different groups each time. What should be concluded?

(A) The structure is strong

(B) Use the largest cluster only

(C) The structure is unstable and should not be acted upon yet

(D) Increase k until the groups agree

7. A clustering is built on past hiring decisions that were themselves biased. What will it most likely do?

(A) Reproduce the same bias in its groups

(B) Remove the bias, since no labels are used

(C) Eliminate protected attributes

(D) Produce random groups

8. Clustering X has a lower inertia than clustering Y, but also a lower silhouette score. What does this tell you?

(A) X is better in every respect

(B) Inertia measures separation, so X is better separated

(C) The two measures always agree, so there is an error

(D) X's clusters are more compact, but the silhouette, which also counts separation, favours Y

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply an unsupervised learning workflow to the following brief: a food-delivery company holds a year of anonymised orders (time, order value, cuisine, delivery area) and wishes to understand the kinds of customer it serves. (10 Marks)

(a) State the preprocessing required and the features you would construct. *[5]*

(b) State the method you would use and how you would validate the result. *[5]*

10. A telecom company wants to act on a customer segmentation and must first establish that it is stable. (10 Marks)

(a) Describe a procedure for testing stability by resampling or by perturbing the data. *[5]*

(b) Explain what result would show that the segments are genuine structure rather than an artefact of one particular sample. *[5]*

----- End of Practice Sheet -----